# ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

**Rockwell Automation PowerFlex 527 (520-series, integrated motion) AC Drive** · Component type: **Embedded Device (EDR) - integrated-motion compact VFD** · Target SL = SL 3

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|---|---|
| **Where to work** | Almost everything is in **Studio 5000 Logix Designer** - the 527 is added as an **EtherNet/IP module / integrated-motion axis** owned by a Logix controller; there is **no local keypad**. Also: **Connected Components Workbench (CCW)** for standalone parameter access, **FactoryTalk Security** for engineering rights, the embedded **EtherNet/IP web page**, firmware via **ControlFLASH / CCW**. |
| **Interfaces** | Dual embedded **EtherNet/IP** only (no DSI/serial control on the 527); configured through the owning Logix controller. Parameters follow the 520-series letter+number scheme but are typically set via the AOP / axis configuration rather than a panel. |
| **Notes** | EDR - 520-series integrated-motion compact drive (PowerFlex 527). Confirm firmware rev and the owning controller. **Important:** like the 520-series generally, the 527 has a light native security stack (no CIP Security, no drive login). Because it is controller-owned, Logix/FactoryTalk Security and segmentation are the primary SL-3 controls - document them explicitly. |
| **How to read** | Read-only: review the Studio 5000 module/axis config, the CCW parameter set, and the embedded web page. Native how-to (exact location/param) + external-tool validation + pass, with honest Partial/Compensating where native controls are thin. Scans authorized; fuzz ISOLATED lab only. |

## FR 1 — Identification & Authentication Control (IAC)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 1.1** | Human user identification & auth | The 527 has **no local login/keypad**; all human access is through Studio 5000 / CCW on the engineering workstation. Enforce authentication there: confirm **FactoryTalk Security** governs who may go online with the owning controller and edit the 527 module/axis, and that CCW/RSLinx access is controlled on the EWS. Record the drive-level absence as a residual. **Test:** confirm no external unauthenticated path (open web/CIP) can change the drive from outside the cell.<br>**Ext. tools:** FactoryTalk Admin Console (users); Nmap/CCW (external write attempt - must fail at the zone). | Access authenticated via FactoryTalk/Logix/EWS; drive-level absence documented. | Partial/Compensating. |
| **SR/CR 1.2** | Software process / device identification | Confirm the 527 is uniquely identified: in Studio 5000 the module has a defined IP/host name and is associated to exactly one owning controller; in CCW read the Group C IP params. Each comm partner has a distinct EtherNet/IP identity.<br>**Ext. tools:** Studio 5000 (module IP/owner); CCW (Group C); Wireshark. | Unique drive identity; single owning controller; partners identified. | |
| **SR/CR 1.3** | Account management | No drive accounts exist; manage FactoryTalk users/roles for the owning-controller project and OS accounts on the EWS - least privilege, no defaults, no shared logins.<br>**Ext. tools:** FactoryTalk Admin Console; EWS local/AD accounts. | Surrounding accounts managed/least-privilege; drive has none by design. | Compensating. |
| **SR/CR 1.4** | Identifier management | Confirm a unique drive identity (EtherNet/IP address + module name in Studio 5000 + nameplate serial); no address/name reuse across drives; unique FactoryTalk user IDs.<br>**Ext. tools:** Studio 5000 / CCW device info; network inventory. | Unique user and device identifiers; no reuse. | |
| **SR/CR 1.5** | Authenticator management | Enforce strong authenticators where they exist - FactoryTalk password policy and EWS OS policy. If the embedded web server or any SMTP/email-on-fault feature is enabled, ensure no unauthenticated write; prefer disabling it (7.6).<br>**Ext. tools:** FactoryTalk password policy; embedded web/SMTP check. | Strong managed authenticators at surrounding layers; unneeded services off. | Compensating. |
| **SR 1.6** | Wireless Access Management | **N/A.** No wireless interface on this device; wireless access is managed at the network layer. | N/A - justification recorded (not assessed as a control). | N/A |
| **SR/CR 1.7** | Strength of password-based authentication | Confirm the password-strength policy (length/complexity/lifetime) meets SL-3. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.5** above (shared steps); confirm this requirement individually and adjudicate it separately. | Strength of password-based authentication verified individually. | |

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 1.8** | Public key infrastructure (PKI) certificates | **N/A at the drive.** The 527 does not support CIP Security / device certificates. Record N/A with justification; integrity/confidentiality is provided by segmentation and by TLS at the SCADA/EWS layer.<br>**Ext. tools:** Confirm no CIP Security capability. | N/A at drive (no PKI support); compensating TLS/segmentation upstream. | N/A |
| **SR/CR 1.9** | Strength of public key-based authentication | Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately. | Strength of public key-based authentication verified individually. | |
| **SR/CR 1.10** | Authenticator feedback | Confirm Studio 5000 / FactoryTalk logon and EWS login mask password entry (drive has no login prompt).<br>**Ext. tools:** Visual verification. | Authenticators obscured at the interfaces that have them. | |
| **SR/CR 1.11** | Unsuccessful login attempts | Enforce lockout at the layers that authenticate (FactoryTalk lockout policy, EWS/AD lockout). The drive has no login to lock; note as residual.<br>**Ext. tools:** FactoryTalk / AD lockout policy; SIEM failed-auth. | Lockout at surrounding layers; drive residual noted. | Compensating. |
| **SR/CR 1.12** | System use notification | No drive banner; enforce the approved use notice on the EWS (GPO) and HMI.<br>**Ext. tools:** EWS GPO / HMI banner review. | Use notice at EWS/HMI (compensating). | Compensating. |
| **SR 1.13** | Access via untrusted networks | Confirm the 527 EtherNet/IP is reachable only from the trusted cell zone behind the cell firewall / Stratix ACLs; remote engineering only via VPN; deny drive access from untrusted networks. Key control for a light-security drive.<br>**Ext. tools:** Nmap (from untrusted side - expect no reach); firewall/ACL review. | Access from untrusted networks denied; remote via VPN. | |

## FR 2 — Use Control (UC)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 2.1** | Authorization enforcement | The 527 is **owned by one Logix controller** - confirm the axis/module is associated to only the authorized controller and connection ownership is exclusive, so no other device can command it; FactoryTalk Security limits who may edit that controller's project. **Test:** attempt to establish a second owning connection - it must be refused (already-owned).<br>**Ext. tools:** Studio 5000 (module ownership/connection); attempt 2nd owner (isolated) - refused. | Exclusive controller ownership; FactoryTalk rights least-privilege. | |
| **SR/CR 2.2** | Wireless Use Control | **N/A.** No wireless interface on this device; wireless access is managed at the network layer. | N/A - justification recorded (not assessed as a control). | N/A |
| **SR/CR 2.3** | Use control for portable/mobile devices | Control the engineering laptop and any USB used for firmware; removable-media policy on the EWS; only authorized projects loaded.<br>**Ext. tools:** Physical inspection; EWS removable-media policy. | Portable/engineering media controlled. | |
| **SR/EDR 2.4** | Mobile code | No general-purpose mobile code on the drive; confirm only drive firmware + parameters run and embedded web content is vendor read-only.<br>**Ext. tools:** CCW / embedded web review. | Only trusted firmware/config runs; no untrusted mobile code. | EDR-specific. |
| **SR/CR 2.5** | Session lock | Confirm Studio 5000 / CCW and EWS sessions time out on inactivity and require re-auth (drive has no session concept).<br>**Ext. tools:** Studio 5000 / EWS timeout config. | Engineering sessions lock/timeout; re-auth required. | Compensating. |
| **SR/CR 2.6** | Remote session termination | Confirm remote engineering (RDP/VPN to the EWS) can be terminated and idle sessions close; the owning CIP connection is exclusive and limited.<br>**Ext. tools:** EWS/VPN session review; Studio 5000 connection review. | Remote sessions terminable on demand and on inactivity. | |
| **SR/CR 2.7** | Concurrent session control | Confirm the owning connection is exclusive and concurrent monitoring connections to the embedded adapter are bounded, preventing conflicting simultaneous control.<br>**Ext. tools:** Studio 5000 / adapter connection review; Wireshark. | Ownership exclusive; concurrent connections bounded/monitored. | |

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 2.8** | Auditable events | The drive keeps a fault log but no user-attributed audit. Capture configuration/firmware changes in **FactoryTalk AssetCentre** (Audit + DR) with the acting user, and forward EWS/FactoryTalk events to the SIEM; read the drive fault log via Studio 5000/CCW.<br>**Ext. tools:** AssetCentre audit; Studio 5000/CCW fault log; SIEM. | Changes audited/attributable via AssetCentre/EWS; drive faults captured. | Compensating. |
| **SR/CR 2.9** | Audit storage capacity | Drive fault log is a small fixed depth; retention via AssetCentre / SIEM - confirm sizing and threshold alerting.<br>**Ext. tools:** AssetCentre DB sizing; SIEM retention. | Events offloaded and retained; alert before capacity. | |
| **SR/CR 2.10** | Response to audit processing failures | Confirm the SIEM/AssetCentre alerts on forwarding/collection failure.<br>**Ext. tools:** SIEM / AssetCentre health alerting. | Audit-failure alerting defined. | |
| **SR/CR 2.11** | Timestamps | The drive has no RTC; event time comes from the owning controller/AssetCentre - confirm those are NTP-synced.<br>**Ext. tools:** NTP / controller time review. | Reliable, synchronized timestamps for drive events. | |
| **SR/CR 2.12** | Non-repudiation | Confirm actions are attributable to a specific user (non-repudiation). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.8** above (shared steps); confirm this requirement individually and adjudicate it separately. | Non-repudiation verified individually. | |
| **EDR 2.13** | Physical diagnostic & test interfaces | Control physical interfaces: keep the drive in a locked cabinet; secure the USB firmware port and Ethernet ports; there is no local keypad-start to disable (no HIM). Disable the embedded web server if unused.<br>**Ext. tools:** Physical inspection; CCW/web (disable unused services). | Physical/diagnostic interfaces controlled; unused services disabled. | EDR-specific. |

## FR 3 — System Integrity (SI)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 3.1** | Communication integrity | The 527 EtherNet/IP is **unauthenticated cleartext CIP** (no CIP Security). Provide integrity/confidentiality by containing it in a segmented cell zone and terminating TLS at the SCADA/EWS layer; confirm no drive CIP traffic crosses a zone boundary in cleartext.<br>**Ext. tools:** Wireshark (confirm cleartext CIP stays in-zone); firewall rule review. | Drive comms contained to the segmented zone; no cleartext CIP across boundaries. | Compensating. |
| **SR/EDR 3.2** | Protection from malicious code (component) | Confirm only Rockwell firmware runs and there is no code-execution path. 520-series firmware lacks the 750-series signed-.dmk guarantee - verify provenance against PCDC and control the update path tightly (2.3/3.10).<br>**Ext. tools:** CCW firmware rev; PCDC provenance check. | Only vendor firmware runs; provenance verified; no code path. | EDR-specific/Partial. |
| **SR/CR 3.3** | Security functionality verification | After any change re-verify: controller ownership, connection-loss action, embedded web-server state, and IP config, against the hardening baseline; compare the Studio 5000 module config / CCW project against the archived version.<br>**Ext. tools:** Studio 5000 / CCW compare vs baseline; Nmap (services off). | Security-relevant settings match baseline after change. | |
| **SR/CR 3.4** | Software & information integrity | Archive the Studio 5000 project + 527 parameters in AssetCentre and enable change detection so unauthorized parameter/firmware change is detected. Compare online-vs-offline.<br>**Ext. tools:** AssetCentre change detection; online-vs-offline compare. | Config integrity monitored; unauthorized change detected. | Compensating. |
| **SR/CR 3.5** | Input validation | Confirm the drive tolerates malformed EtherNet/IP/CIP frames without faulting or unsafe motion. **ISOLATED only:** fuzz the adapter on a bench drive and watch the fault log / connection status.<br>**Ext. tools:** ENIP/CIP fuzzer (ISOLATED); Wireshark. | No crash or unsafe motion on malformed input. | Do NOT fuzz production. |
| **SR/CR 3.6** | Deterministic output | Verify the connection-loss / fault reaction so the motor reaches a defined safe state: in Studio 5000 set the module/axis **connection-loss action** (and the drive comm-fault action) to Fault/Stop as designed - not run-through - and where a hardwired safety/enable circuit is fitted, function-test it.<br>**Ext. tools:** Studio 5000 (connection-loss/axis fault action); hardwired enable test (lab). | Drive faults/stops to a defined safe state on connection loss. | |

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 3.7** | Error handling | Confirm graceful fault handling: review the fault log and disable auto-restart unless justified so errors do not cause unsafe restart.<br>**Ext. tools:** Studio 5000 / CCW (fault log; auto-restart config). | Graceful fault handling; no unsafe restart. | |
| **SR/CR 3.8** | Session integrity | **N/A at drive (no secured session).** Legacy CIP sessions are unauthenticated; integrity is provided by exclusive ownership + containment. Record N/A + compensating segmentation.<br>**Ext. tools:** Wireshark (confirm in-zone containment). | N/A at drive; compensating ownership/segmentation recorded. | N/A |
| **SR/CR 3.9** | Protection of audit information | Protect the AssetCentre audit DB and SIEM store; restrict who can read/clear the drive fault log.<br>**Ext. tools:** AssetCentre / SIEM access review. | Audit records protected from unauthorized read/modify/delete. | |
| **EDR 3.10** | Support for updates | Update firmware via ControlFLASH / CCW with firmware from PCDC; verify revision/provenance; plan the update in a maintenance window with rollback to the prior known-good file. (No on-drive signature check like 750-series.)<br>**Ext. tools:** ControlFLASH/CCW; PCDC checksum/provenance. | Controlled updates from verified source, with rollback. | EDR-specific/Partial. |
| **EDR 3.11** | Physical tamper resistance/detection | Keep the drive in a locked, sealed cabinet; control the USB/Ethernet ports; AssetCentre change records and the drive fault log provide change evidence.<br>**Ext. tools:** Physical inspection; AssetCentre / fault-log review. | Cabinet locked/sealed; interfaces controlled; changes logged. | EDR-specific. |
| **EDR 3.12** | Provisioning product-supplier roots of trust | **N/A at drive.** 520-series provides no signed-boot supplier root of trust. Record N/A; compensate via firmware update-provenance discipline (3.10), physical control, and segmentation.<br>**Ext. tools:** Confirm no secure-boot / signed-firmware root capability. | N/A at drive (no supplier RoT); compensating controls recorded. | N/A |
| **EDR 3.13** | Provisioning asset-owner roots of trust | **N/A at drive.** 527 provides no CIP Security / asset-owner certificate provisioning. Record N/A; asset-owner trust is enforced at the controller/SCADA/EWS layer and by segmentation.<br>**Ext. tools:** Confirm no CIP-Security / certificate-provisioning capability. | N/A at drive (no asset-owner RoT); compensating controls recorded. | N/A |
| **EDR 3.14** | Integrity of the boot process | **N/A at drive (no verified/secure boot).** Record N/A; mitigate via controlled firmware provenance (3.10), physical control, and monitoring.<br>**Ext. tools:** Confirm no secure-boot capability; CCW firmware rev. | N/A at drive; compensating provenance/physical controls recorded. | N/A |

## FR 4 — Data Confidentiality (DC)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 4.1** | Information confidentiality | Confirm information confidentiality — sensitive data is protected in transit and at rest. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.1** above (shared steps); confirm this requirement individually and adjudicate it separately. | Information confidentiality verified individually. | |
| **SR/CR 4.2** | Information persistence | On decommission/repurpose, in CCW/Studio 5000 reset the drive to factory defaults and clear config so it is not recoverable.<br>**Ext. tools:** CCW/Studio 5000 (reset to defaults). | Sensitive configuration purged on decommission or reuse. | |
| **SR/CR 4.3** | Use of cryptography | **N/A at drive.** No native crypto on the 527; disable the embedded web server / any SMTP feature and rely on TLS at the SCADA/EWS layer and segmentation.<br>**Ext. tools:** Confirm no drive crypto; disable web/SMTP (7.6). | N/A at drive; compensating TLS/segmentation upstream. | N/A |

## FR 5 — Restricted Data Flow (RDF)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 5.1** | Network segmentation | Confirm the drive sits in a controlled cell/zone behind a firewall / Stratix ACLs; only required EtherNet/IP flows cross the boundary; no flat convergence with IT. Primary control for this light-security drive.<br>**Ext. tools:** Nmap by zone; firewall/ACL rule review. | Drive in a segmented zone; only required flows cross the boundary. | |
| **SR 5.2** | Zone boundary protection | Confirm zone-boundary protection — only required flows cross the boundary (system SR / network device).<br>**Steps:** for this product this is assessed with the same evidence as **SR/CR 5.1** above (shared steps); confirm this requirement individually and adjudicate it separately. | Zone boundary protection verified individually. | |
| **SR 5.3** | General-purpose person-to-person comms restrictions | Largely **N/A**, but if the embedded adapter's **fault email/SMTP notification** is enabled, disable it or restrict SMTP so the drive originates no general-purpose messaging.<br>**Ext. tools:** CCW / embedded web (SMTP/email config) - confirm disabled. | N/A / any email-notification feature disabled. | N/A |
| **SR 5.4** | Application partitioning | **N/A.** Single-purpose drive; no separable applications to partition.<br>**Ext. tools:** Confirm single-function device. | N/A - justification recorded. | N/A |

## FR 6 — Timely Response to Events (TRE)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 6.1** | Audit log accessibility | Confirm drive faults (Studio 5000/CCW fault log) and AssetCentre change records are readable/exportable and forwarded to the SIEM; the cell is passively monitored (OT IDS on a mirror port).<br>**Ext. tools:** AssetCentre export; passive OT IDS (mirror port). | Events accessible/exportable and monitored in near-real-time. | |
| **SR/CR 6.2** | Continuous monitoring | Confirm continuous monitoring — events forwarded to and monitored by a SIEM/OT-IDS in near-real-time.<br>**Steps:** for this product this is assessed with the same evidence as **SR/CR 6.1** above (shared steps); confirm this requirement individually and adjudicate it separately. | Continuous monitoring verified individually. | |

## FR 7 — Resource Availability (RA)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 7.1** | Denial-of-service protection | Confirm DoS resilience: the adapter CIP connection limit plus cell-firewall rate-limiting prevent a flood from denying control; monitor the owning connection and comms status while testing.<br>**Ext. tools:** Adapter connection review; traffic generator (ISOLATED). | Resources bounded; drive control continues under load. | |
| **SR/CR 7.2** | Resource management | Confirm the adapter bounds resource use (fixed CIP connection limit) so a single service cannot starve control.<br>**Ext. tools:** Adapter connection/limits review. | Resources bounded; no exhaustion-based loss of control. | |
| **SR/CR 7.3** | Control system backup | Confirm the Studio 5000 project / 527 parameters are backed up (AssetCentre DR, versioned, offline) and a restore is tested for a known-good rebuild.<br>**Ext. tools:** AssetCentre DR review; restore test (lab). | Backups current and recovery validated. | |
| **SR/CR 7.4** | Control system recovery & reconstitution | Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately. | Control system recovery & reconstitution verified individually. | |
| **SR/CR 7.5** | Emergency Power | **N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer. | N/A - justification recorded (not assessed as a control). | N/A |
| **SR/CR 7.6** | Network & security configuration settings | Harden per baseline: disable the embedded **web server** and any **email/SMTP** feature if unused, set the connection-loss action, confirm exclusive controller ownership, and disable unused services/ports.<br>**Ext. tools:** Studio 5000/CCW/web (disable web/SMTP); Nmap (exposed services). | Least functionality; unused services disabled per baseline. | |

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / firmware |
|---|---|---|---|---|
| **SR/CR 7.7** | Least functionality | Confirm least functionality — unused services/ports/protocols are disabled. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.6** above (shared steps); confirm this requirement individually and adjudicate it separately. | Least functionality verified individually. | |
| **SR/CR 7.8** | Component inventory | Confirm the drive (catalog, firmware rev, adapter) is inventoried so vulnerabilities track against Rockwell Automation security advisories.<br>**Ext. tools:** Studio 5000/CCW/AssetCentre inventory; ReadOnlyEnum. | Accurate component/firmware inventory maintained. | |

## External Test Tools — how to use

| Tool / method | Verifies | How to use — steps / example | Constraints |
|---|---|---|---|
| **Studio 5000 Logix Designer** | Owner/axis config & native review (all FRs) | Read the 527 module/axis: owning controller, IP/name, connection-loss action, fault log, firmware rev; compare to baseline. | Read/authorized project copy. |
| **CCW** | Standalone parameter access (1.2, 3.x, 4.2, 7.6) | Read Group C IP, comm-fault actions, and firmware rev when not viewing through Logix. | Read/authorized project copy. |
| **FactoryTalk Security / Admin Console** | Identity & authorization (FR1, 2.1) | Read FactoryTalk users/roles for the owning-controller project; least-privilege check. | Read-only. |
| **FactoryTalk AssetCentre** | Audit, change detection & DR (2.8, 3.4, 6.x, 7.3) | Audit config/firmware changes and provide versioned backup/restore. | Non-intrusive. |
| **ControlFLASH / PCDC** | Firmware update & provenance (3.2, 3.10) | Update from verified PCDC files; confirm revision/provenance. | Maintenance window. |
| **Nmap / ReadOnlyEnum / Wireshark** | Exposed-service, zone & comms review (1.13, 3.1, 5.x, 7.6) | Enumerate EtherNet/IP + embedded web per zone; confirm cleartext CIP stays in-zone. | Read-only/passive; authorized. |
| **ENIP/CIP fuzzer / traffic generator** | Input validation & DoS resilience (3.5, 7.1) | Malformed-frame / high-rate EtherNet/IP testing on a bench drive while watching fault log / connection health. | ISOLATED/non-production; written authorization. |

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.